

Samagra Bharti

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EDUCATION

International Institute of Information Technology, Hyderabad <i>B.Tech in Computer Science & M.S. (Research) in Computing and Human Sciences</i>	October 2022 – Present CGPA: 8.21
Rajendra Vidyalaya, Jamshedpur <i>ICSE (Class 10): 97.6% — ISC (Class 12): 97%</i>	2018 – 2022

EXPERIENCE

Teradata <i>AI Platform Software Engineer Intern</i>	May 2026 – Present Hyderabad, India
- Built a full-stack AI agent optimizing SIT/UAT regression testing via a Jaccard scoring engine and pgvector persistence, targeting a 50–65% reduction in total CI cycle time across 130+ suites. - Designed a multi-stage repository onboarding pipeline with diff-scoped LLM analysis to scale token costs with change size; supported direct manifest bypass via configuration files. - Delivered an agent platform with a React 19 frontend, Kubernetes container orchestration, and Prometheus tracking; traced 97% of redundant staging failures to a single core root cause.	
Human Science Research Centre - IIIT <i>Undergraduate Researcher for Mapping the Himalayas under Dr. Aniket Alam</i>	May 2024 – Present Honors Research
- Deployed a production historical research web service on AWS using a modular MERN architecture to orchestrate HTTP request traffic; platform code at highlandhistories.org. - Built an automated NLP pipeline converting trade text corpora into structured data artifacts via text extraction, NER, entity co-occurrence mining, and subject-relation-object extraction math. - Generated analytical knowledge graphs and interactive HTML network visualizations for Himalayan trade paths; engineered domain-adaptive BERT-BiLSTM sequence taggers for custom entity labels.	
Cart Genie <i>Software Engineer Intern</i>	January 2024 – May 2024 Hyderabad, India
- Engineered asynchronous background processing routines for real-time delivery message parsing on mobile clients. - Built optimized cross-platform UI views, improving client rendering efficiency and interaction smoothness. - Enforced architectural guidelines and test coverage benchmarks through git-driven collaborative code reviews.	

PROJECTS

Unix Shell <i>C, POSIX, Network Programming, OS, Git, MakeFile</i>
Engineered a modular UNIX shell, implemented full OS-level features like command history, I/O redirection, pipes, background and foreground processes, signals and custom commands for directory management and processing information.
Network File System <i>TCP/IP, Socket Programming, C, Multithreading, Git, MakeFile</i>
Developed a client-server distributed file system in C with a centralized Naming Server for metadata and request management, leveraging TCP/IP sockets, concurrent client handling, OS-level synchronization, and Linux system calls to ensure consistency, fault tolerance, and efficient file operations.
Bridging Language Models and Math Word Problems <i>RoBERTa, Bi-LSTM, GTS, NLP, PyTorch</i>
Developed and evaluated RoBERTa-based MWP solvers on Math23K using Goal-Driven Tree Decoders and RPKHS, achieving 71.12% answer accuracy, 63.28% equation accuracy for symbolic mathematical reasoning and equation parsing.
Pixel Based Classification Study <i>Remote Sensing, Ensemble, ML, GEE, Python</i>
Built a Google Earth Engine-based weighted ensemble of RF, SVM, CART, and XGBoost models for land-cover classification on Sentinel-2, LISS-4, and Landsat-8 imagery, with applications in geospatial asset monitoring and structured data modeling.
Spider-Verse Mini-World Database <i>Python, SQL</i>
Designed a relational database to manage Spider-Verse characters, missions, and villains with complex and recursive associations; developed SQL queries for analytics and operations.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Flutter, MySQL, JavaScript, HTML/CSS, Java, Bash
Frameworks & Libraries: React, Node.js, Flask, JUnit, WordPress, CUDA, AWS
Tools & Platforms: Linux/ Unix, Git, Google Earth Engine, Nginx, VS Code, Android Studio, PyTorch, TensorFlow, QGIS, MongoDB
Relevant Coursework: DSA, Linear Algebra, Algorithmic Analysis & Design, Probability & Statistics, Operating System & Networks, NLP

ACHIEVEMENTS, LEADERSHIP & EXTRACURRICULARS

Hackathon Finalist: Engineered a peninsular cyclone modeling framework at Megathon 24 using Earth Engine assets.
Dean's List: Recognized for high academic standing throughout the Monsoon 24 academic term.
Member of Students Parliament: Elected governance official serving student council action and policy briefs.
Pianist: Active student and practitioner of classical and modern piano arrangements for over 16 years.